

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 9

- (1) Suppose that $T : V \rightarrow V$ is an F -linear transformation with $\dim_F V = n$. Choose a basis $\mathcal{B}$ for V , and set $A = [T]_{\mathcal{B}}$. Show that the characteristic polynomial $\text{ch}_A(x)$ does not depend on which basis you chose.

Solution. If the basis is $\mathcal{B} = (v_1, \dots, v_n)$ and if $\mathcal{B}' = (v'_1, \dots, v'_n)$ is a different basis, with $C = [c_{i,j}]$ the change of basis matrix with

$$v'_j = \sum_i c_{i,j} v_i$$

then we have $A = [T]_{\mathcal{B}} = C[T]_{\mathcal{B}'}C^{-1}$. Therefore, if $A' = [T]_{\mathcal{B}'}$, then we have $A' = C^{-1}AC$

$$\text{ch}_{A'}(x) = \det(A' - xI_n) = \det(C^{-1}AC - xI_n) = \det(C^{-1}(A - xI_n)C).$$

Here, in the last equality we are using the fact that $C^{-1}(xI_n)C = xC^{-1}I_nC = xI_n$.

But we then have

$$\det(C^{-1}(A - xI_n)C) = \det(C^{-1}) \det(A - xI_n) \det(C) = \det(C)^{-1} \text{ch}_A(x) \det(C) = \text{ch}_A(x).$$

Definition 1. If $\lambda \in F$ and $A \in M_{n \times n}(F)$, the λ -**eigenspace** for A $E_{\lambda}(A)$ is the kernel of $A - \lambda I_n$. The **geometric multiplicity** of λ (with respect to A) is the dimension of $E_{\lambda}(A)$

- (2) Find the eigenvalues of

$$A = \begin{bmatrix} 7 & 4 & 4 \\ 0 & -1 & 0 \\ -8 & -4 & -5 \end{bmatrix} \in M_{3 \times 3}(\mathbb{C}).$$

Compute bases for its eigenspaces, and then diagonalize A or explain why this is not possible.

Solution. The characteristic polynomial is

$$\det \begin{bmatrix} 7-x & 4 & 4 \\ 0 & -1-x & 0 \\ -8 & -4 & -5-x \end{bmatrix} = 0$$

Expanding along the second row:¹

$$\begin{aligned} \text{ch}_A(x) &= (-1-x) [(7-x)(-5-x) - (4)(-8)] \\ &= -(x+1) [x^2 - 2x - 35 + 32] \\ &= -(x+1)(x^2 - 2x - 3) \\ &= -(x+1)^2(x-3) \end{aligned}$$

The eigenvalues are $\lambda_1 = -1$ (algebraic multiplicity 2) and $\lambda_2 = 3$ (algebraic multiplicity 1).

To compute the eigenspaces E_{-1} and E_3 , we need to compute kernels. **For $\lambda = -1$:** We solve $(A + I_3)v = 0$:

$$\begin{bmatrix} 8 & 4 & 4 \\ 0 & 0 & 0 \\ -8 & -4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

¹We are using the formula for this from problem 6 below to simplify things.

The system reduces to $2x + y + z = 0$. A basis for the eigenspace E_{-1} is:

$$\mathcal{B}_{-1} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

For $\lambda = 3$: We solve $(A - 3I_3)v = 0$:

$$\begin{bmatrix} 4 & 4 & 4 \\ 0 & -4 & 0 \\ -8 & -4 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This implies $y = 0$ and $x + z = 0$. A basis for the eigenspace E_3 is:

$$\mathcal{B}_3 = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Since we clearly have 3 linearly independent eigenvectors between the two bases, we see that the matrix is diagonalizable. Explicitly, if we take the combined basis to be

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\},$$

then this is an eigenbasis for A , and we have

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -2 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ -2 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}^{-1}.$$

(3) Find the eigenvalues of

$$B = \begin{bmatrix} -1 & 0 & -3 & 0 \\ 3 & 5 & 3 & -3 \\ 0 & 0 & 2 & 0 \\ 3 & 6 & 3 & -4 \end{bmatrix} \in M_{4 \times 4}(\mathbb{C}).$$

Compute bases for its eigenspaces, and then diagonalize B or explain why this is not possible.

Solution. Expanding along the third row of $B - xI_4$, we get

$$\begin{aligned} \text{ch}_B(x) &= (2-x)(-1-x)[(5-x)(-4-x) - (-3)(6)] \\ &= (2-x)(-1-x)(x^2 - x - 2) \\ &= (2-x)(-1-x)(x-2)(x+1) \\ &= (x-2)^2(x+1)^2 \end{aligned}$$

For $\lambda = 2$: Solving $(B - 2I_4)v = 0$ leads to the row-reduced system:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{v} = \mathbf{0}$$

A basis for the eigenspace E_2 is:

$$\mathcal{B}_2 = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

For $\lambda = -1$: Solving $(B + I_4)v = 0$ leads to the row-reduced system:

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{v} = \mathbf{0}$$

A basis for the eigenspace E_{-1} is:

$$\mathcal{B}_{-1} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Again, these combine to give an eigenbasis, and we have

$$B = \begin{bmatrix} -1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}^{-1}$$

Definition 2. If $\lambda \in F$ and $A \in M_{n \times n}(F)$, the **algebraic multiplicity** of λ (with respect to A) is the largest power of $x - \lambda$ that divides $\text{ch}_A(x)$.

- (4) Find a *non*-diagonalizable matrix $A \in M_3(\mathbb{R})$ such that, for all $\lambda \in \mathbb{R}$, the geometric multiplicity of λ equals the algebraic multiplicity of λ .

Solution. Take the matrix

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The first 2×2 -block has characteristic polynomial $x^2 + 1$, and you can use this to see that $\text{ch}_A(x) = (x^2 + 1)(x - 1)$. This means that the only eigenvalue A has in $\mathbb{R}$ is 1 with algebraic and geometric multiplicity both equal to 1. However, the matrix is *not* diagonalizable over $\mathbb{R}$ because the other two zeros of the characteristic polynomial only show up in $\mathbb{C}$.

- (5) Fix $(n - 1)$ -vectors $v_1, \dots, v_{n-1} \in F^n$ and $i \in \{1, \dots, n\}$. Consider the function

$$S_{(v_1, \dots, v_{n-1})}^i : F^n \rightarrow F$$

$$w \mapsto \det([v_1 \ v_2 \ \cdots \ v_{i-1} \ w \ v_i \ \cdots \ v_{n-1}])$$

In other words, we are fixing all the columns of a matrix except the i -th column, and viewing the determinant as a function of that column.

- (a) Show that $S_{(v_1, \dots, v_{n-1})}^i$ is an F -linear transformation, and conclude that there exist elements

$$C_{i,j}(v_1, \dots, v_{n-1}) \in F$$

for $j = 1, \dots, n$ such that

$$S_{(v_1, \dots, v_{n-1})}^i \left(\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \right) = C_{i,1}(v_1, \dots, v_{n-1})a_1 + C_{i,2}(v_1, \dots, v_{n-1})a_2 + \cdots + C_{i,n}(v_1, \dots, v_{n-1})a_n$$

Solution. The fact that $S_{(v_1, \dots, v_{n-1})}^i$ is a linear transformation is a consequence of the linearity property of the determinant. Explicitly, what we are saying is that

$$S_{(v_1, \dots, v_{n-1})}^i(a_1 w_1 + a_2 w_2) = \det([v_1 \ v_2 \ \cdots \ v_{i-1} \ a_1 w_1 + a_2 w_2 \ v_i \ \cdots \ v_{n-1}]) = a_1 \det([v_1 \ v_2 \ \cdots \ v_{i-1} \ w_1 \ v_i \ \cdots \ v_{n-1}]) + a_2 \det([v_1 \ v_2 \ \cdots \ v_{i-1} \ w_2 \ v_i \ \cdots \ v_{n-1}])$$

Since $S_{(v_1, \dots, v_{n-1})}^i$ is a linear transformation from F^n to F , it is given by an $n \times 1$ -matrix

$$[C_{i,1}(v_1, \dots, v_{n-1}) \ C_{i,2}(v_1, \dots, v_{n-1}) \ \cdots \ C_{i,n}(v_1, \dots, v_{n-1})]$$

- (b) By taking $w = e_j$, show that $C_{i,j}(v_1, \dots, v_{n-1})$ depends only the $(n-1) \times (n-1)$ -matrix obtained by taking the $n \times (n-1)$ matrix

$$A = [v_1 \ \cdots \ v_{n-1}]$$

and removing the j -th row: In other words, $C_{i,j}$ is a certain function on $(n-1) \times (n-1)$ -matrices.

Solution. If we take $w = e_j$, then we see that

$$S_{(v_1, \dots, v_{n-1})}^i(e_j) = C_{i,j}.$$

On the other hand, if

$$A = \begin{bmatrix} a_{1,1} & \cdots & a_{1,n-1} \\ \vdots & \ddots & \vdots \\ a_{j-1,1} & \cdots & a_{j-1,n-1} \\ a_{j,1} & \cdots & a_{j,n} \\ a_{j+1,1} & \cdots & a_{j+1,n-1} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n-1} \end{bmatrix}$$

we can use column operations on the matrix

$$[v_1 \ \cdots \ v_i \ e_j \ v_{i+1} \ \cdots \ v_{n-1}] = \begin{bmatrix} a_{1,1} & \cdots & 0 & \cdots & a_{1,n-1} \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ a_{j-1,1} & \cdots & 0 & \cdots & a_{j-1,n-1} \\ a_{j,1} & \cdots & 1 & \cdots & a_{j,n} \\ a_{j+1,1} & \cdots & 0 & \cdots & a_{j+1,n-1} \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ a_{n,1} & \cdots & 0 & \cdots & a_{n,n-1} \end{bmatrix}$$

to get rid of all the non-diagonal entries in the j -th column using the 1 in the middle, and see that the determinant is the same as that of the matrix

$$\begin{bmatrix} a_{1,1} & \cdots & 0 & \cdots & a_{1,n-1} \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ a_{j-1,1} & \cdots & 0 & \cdots & a_{j-1,n-1} \\ 0 & \cdots & 1 & \cdots & 0 \\ a_{j+1,1} & \cdots & 0 & \cdots & a_{j+1,n-1} \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ a_{n,1} & \cdots & 0 & \cdots & a_{n,n-1} \end{bmatrix}$$

Therefore, $C_{i,j}$ only depends on the matrix obtained by removing the j -th row from A , and can be viewed in this way as a function on $(n-1) \times (n-1)$ -matrices. More precisely, given an $(n-1) \times (n-1)$ -matrix

$$B = \begin{bmatrix} b_{1,1} & \cdots & b_{1,n-1} \\ \vdots & \ddots & \vdots \\ b_{j-1,1} & \cdots & b_{j-1,n-1} \\ b_{j,1} & \cdots & b_{j,n-1} \\ \vdots & \ddots & \vdots \\ b_{n-1,1} & \cdots & b_{n-1,n-1} \end{bmatrix}$$

we take the determinant of

$$\begin{bmatrix} b_{1,1} & \cdots & 0 & \cdots & b_{1,n-1} \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ b_{j-1,1} & \cdots & 0 & \cdots & b_{j-1,n-1} \\ 0 & \cdots & 1 & \cdots & 0 \\ b_{j,1} & \cdots & 0 & \cdots & b_{j,n-1} \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ b_{n-1,1} & \cdots & 0 & \cdots & b_{n-1,n-1} \end{bmatrix}$$

- (6) Consider the function $C_{i,j}$ from the previous problem, and think of it as a function on $(n-1) \times (n-1)$ -matrices.

- (a) Show that $C_{i,j}$ satisfies all the properties of the $(n-1) \times (n-1)$ determinant *except* that $C_{i,j}(I_{n-1}) = (-1)^{i+j}$.

Solution. The fact that the function satisfies the properties of the determinant with respect to column operations is immediate from the definition. The last property amounts to computing the determinant of the matrix obtained by taking $I_{n-1,n-1}$ and adding an extra column between the $(i-1)$ -th and i -th columns and an extra row between the $(j-1)$ -th and j -th rows, and taking all the extra entries *except* the (j,i) -th entry to be 0, and taking the (j,i) -th entry to be 1. For instance, if $n = 4$, $j = 2$ and $i = 3$, then we are looking at the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and if $n = 5$, with $j = 4$ and $i = 2$, we get the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The number of column swaps we have to make to get the matrix *back* to the identity is the absolute value $|i-j|$. In the first example, it's just 1, and in the second it takes 2 swaps. This tells us that the determinant is $(-1)^{i-j} = (-1)^{i+j}$ (this is because even powers of -1 don't make a difference).

- (b) Use this to conclude that, for any square matrix $A \in M_{n \times n}(F)$, we have

$$\det(A) = (-1)^{i+1} (a_{i,1}|A_{i,1}| - a_{i,2}|A_{i,2}| + \cdots + (-1)^{j-1} a_{i,j}|A_{i,j}| + \cdots + (-1)^{n-1} a_{i,n}|A_{i,n}|)$$

where $A_{i,j}$ is as in Lecture 26.

Solution. By problem 5(a), taking $v_1, \dots, v_{n-1}$ to be the columns of A *except* the i -th column, and taking w to be the i -th column of A , we find

$$\det(A) = C_{i,1}(v_1, \dots, v_{n-1})a_{1,i} + C_{i,2}(v_1, v_2, \dots, v_{n-1})a_{2,i} + \cdots + C_{i,n}(v_1, \dots, v_{n-1})a_{n,i}.$$

By part (a), and the uniqueness of the determinant, we see that $(-1)^{i+j} C_{i,j}$ is the determinant function on $(n-1) \times (n-1)$ -matrices. More precisely, if we take $A_{j,i}$ to be the matrix obtained by removing the j -th row and i -th column of A , then we obtain

$$(-1)^{i+j} C_{i,j}(v_1, \dots, v_{n-1}) = \det(A_{j,i}).$$

This tells us that

$$\det(A) = (-1)^{i+1} a_{1,i}|A_{1,i}| + (-1)^{i+2} a_{2,i}|A_{2,i}| + \cdots + (-1)^{i+n} a_{n,i}|A_{n,i}|.$$

This looks *almost* like what we want except that the order of indices is flipped. But this is fixed by just noting that $\det(A) = \det(A^T)$, and the formula for $\det(A^T)$ is obtained from that of $\det(A)$ by simply flipping all the indices.

Definition 3. The **adjugate matrix** $\text{adj}(A)$ is the matrix whose (i, j) -th entry is $(-1)^{i+j}|A_{i,j}|$.

- (7) Show that $A \cdot \text{adj}(A) = \det(A)I_n$. Use this to give a different proof that A is invertible if and only if $\det(A) \neq 0$, and give a formula for A^{-1} in terms of the adjugate matrix.

Solution. There is a *typo* here: The adjugate matrix should have had its (i, j) -th entry given by $(-1)^{i+j}|A_{j,i}|$. The equality from problem 6(b) tells us that the (i, i) -th entry of $A\text{adj}(A)$ is exactly $\det(A)$.

To finish we must know that if $i \neq j$, then the (i, j) -th entry of the product is 0. But this entry is

$$a_{i,1}(-1)^{j+1}|A_{j,1}| + a_{i,2}(-1)^{j+2}|A_{j,2}| + \cdots + a_{i,n}(-1)^{j+n}|A_{j,n}|$$

If we replace A with its transpose matrix and take $b_{i,j} = a_{j,i}$ and $B_{i,j} = A_{j,i}$, then we can rewrite this entry as

$$b_{1,i}(-1)^{j+1}|B_{1,j}| + b_{2,i}(-1)^{j+2}|B_{2,j}| + \cdots + b_{n,i}(-1)^{j+n}|B_{n,j}|.$$

Now, if $v_1, \dots, v_{j-1}, v_j, \dots, v_n$ are the columns of A^T *except* for the j -th column, then by 6(a) we have

$$(-1)^{j+r}|B_{r,j}| = C_{j,r}(v_1, \dots, v_{j-1}, v_j, \dots, v_n),$$

and so going back to the definition in problem 5, we see that we have

$$\begin{aligned} & b_{1,i}(-1)^{j+1}|B_{1,j}| + b_{2,i}(-1)^{j+2}|B_{2,j}| + \cdots + b_{n,i}(-1)^{j+n}|B_{n,j}| \\ = & b_{1,i}C_{j,1}(v_1, \dots, v_{j-1}, v_j, \dots, v_n) + b_{2,i}C_{j,2}(v_1, \dots, v_{j-1}, v_j, \dots, v_n) + \cdots + b_{n,i}C_{j,n}(v_1, \dots, v_{j-1}, v_j, \dots, v_n) \\ & = \det([v_1 \ v_2 \ \cdots \ v_{j-1} \ v_i \ v_j \ \cdots \ v_n]) \end{aligned}$$

Since $i \neq j$, this means that v_i appears *twice* in the columns of this matrix, and so the determinant vanishes, as desired.

This expression for the inverse is called Cramer's rule.